

Hybrid power system options for off-grid rural electrification in northwestern region of Nigeria

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Abstract

Electrical power supply is inadequate in Nigeria, and most remote settlements in the country are not connected to the national grid. This study, therefore, investigates a set of possible off-grid hybrid power systems that can meet the electrical load of a model rural community of 100 households in Kaduna and Katsina States in the northwestern region of Nigeria at the lowest energy cost. Two off-grid hybrid power systems and a stand-alone diesel generator are modeled, simulated, and analyzed for the two locations with different solar and wind energy resources using Hybrid Optimization Model for Electric Renewable (HOMER Pro 3.7.5) software. The optimization results show that a PV-wind-diesel hybrid power system is sufficient to meet the load demand of the two locations at the least cost. The net present cost (NPC) and cost of energy (COE) for the optimal system are US\$435,818 and 0.203 US\$/kWh in Kaduna and US\$322,091 and 0.150 US\$/kWh in Katsina. Thus, the optimal system in Katsina has lower NPC than that in Kaduna. In comparison to the grid tariff in Nigeria, the reported cost of energy in this study is cost effective for rural settlements that are far from the existing grid. Thus, the hybrid PV-wind-diesel system is recommended for Kaduna and Katsina. The adoption of this hybrid system in the study locations will boost the social, educational, and economic activities in the villages leading to better education, improved healthcare delivery and enhancement of local businesses.

Keywords: *hybrid power system, simulation, solar, wind, HOMER, electricity*

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1. Introduction

Nowadays, the availability of electrical power has become vital for human well-being and the socio-economic development of a society. The shortage of electricity is the most pronounced problem in most African countries. According to the International Energy Agency (IEA) report [1], around 600 million people lack access to electricity in these countries. Irregular and unstable electricity supply is one of the major challenges in Nigeria. As a country with over 200 million people, the total installed electricity generation capacity from hydro- and gas-power infrastructures is about 13,700 MW, while the available electricity generation capacity at the last quarter of year 2023 was around 4,922.26 MW [2]. GIZ-NESP [3] noted that the inefficiency of the power sector usually resulted in power shortages leaving households and business enterprises without reliable electricity frequently. Another report by the IEA [4] shows that the overall electrification rate in Nigeria is 58%. This implies that about 42% of the populace has no access to electricity supply. The report also indicates that access to electricity in rural areas is about 39%. There are many rural villages without access to electricity in Nigeria. In these villages, a few affluent people among the villagers use petrol or diesel generators for electrification, while the majority of the population rely on kerosene lanterns and traditional biomass fuels, such as firewood,

for lighting. Grid extension to some of the un-electrified rural villages is unprofitable because of their long distance from the nearest grid, difficult terrain, and the sparsely populated nature of the villages. Oparaku's [5] report on rural electrification in Nigeria revealed that the breakeven points of small and large photovoltaic (PV) energy systems with grid extension are respectively at distances of about 26 and 48 km from the existing grid. The author therefore advocates PV systems as a cost-effective option for rural communities with low electrical energy demand. Szabo et al.'s [6] study on electrification options for rural Africa revealed that off-grid solar option is more economical for rural populations who live within the 100 km boundaries of the existing grid.

Renewable energy (RE) from solar, wind, and hydropower has been widely used as an alternative power source for rural settlements far from the existing grid. Vincent-Akpu [7] alluded that RE could meet a significant portion of our energy need with fewer harmful effects on the environment if it is properly harnessed. Nema et al. [8] and Bagiorgas et al. [9] observed that, among the various RE resources, wind and solar energy have proved to be very viable options for electricity generation, and as such among the fastest growing energy resources in the world. Nigeria

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possesses abundance of untapped wind and solar energy resources that could be harnessed for power generation, especially in the remote villages. Studies show that the northern part of the country has substantial solar energy resources which could be complemented by wind for hybrid renewable power system operation [10–12]. Harnessing energy from the renewable resources in Kaduna and Katsina through a hybrid power system will provide cheap, clean, and reliable energy for lighting and powering electrical appliances in the remote villages.

Saheb-Koussa et al. [13] reported that the hybrid power system is more reliable for electrifying remote settlements than using a PV or a wind power system separately. Colle et al. [14] and Saheb-Koussa et al. [15] also stated that the hybrid power system is less expensive than using only a diesel power system. Hybrid renewable power systems (HRPSs) are globally accepted as feasible solutions for rural electrification [16]. An HRPS is adjudged as the best option for rural electrification in locations where grid extension is not economically viable [17]. Tesema and Bekele [18] evaluated the techno-economic feasibility of different HRPSs for a remote rural village in the Somali region of Ethiopia using Hybrid Optimization Model for Electric Renewable (HOMER Pro 3.7.5) software. Their findings indicate that the wind-PV-diesel generator hybrid system was the most viable system for the village. Chisale and Sari [19] simulated a stand-alone HRPS for the Dwangwa area of Malawi using the HOMER software. The simulation results show that the feasible option for the village is a hydropower-wind-solar-battery system. Also, Sabed-Koussa et al. [13] simulated hybrid power options for four rural villages in Algeria using the HOMER software. The study revealed that the energy cost depends largely on the renewable energy potential of the selected sites. In another study, Zailan et al. [20] investigated the feasibility of using an off-grid PV-wind-diesel energy systems for some coastal areas in Pekan, Pahang, Malaysia. The techno-economic analysis of the study conducted using the HOMER software indicated that the PV-wind-diesel energy system has a total net present cost of US\$61, 911 with a cost of energy of US\$0.66/kWh.

A critical survey of the existing literature shows that HRPSs have been simulated and installed for many rural locations in Africa and other regions across the world. However, the use of HRPSs for rural electrification in Nigeria is not well pronounced. Solar

power system is majorly used for remote rural electrification projects in Nigeria despite the abundant wind energy resources in some northwestern states of Nigeria. Thus, this study aims to determine the optimal hybrid power system for remote rural settlements at the lowest energy cost in two selected states (Kaduna and Katsina) in northwest Nigeria. The study also attempts to compare the cost of generating electricity from diesel generator with the various possible off-grid energy system configurations. The result of the research will be useful to the government, researchers, wind and solar energy system investors, and other stakeholders in making informed decisions about wind and solar energy applications in the region.

1.1. Study area

The northwestern region of Nigeria comprises seven states which lie between longitudes 3.5° and 11° E, and latitudes 8° and 14° N as shown in **Figure 1**. The region has an estimated population of 45 million people, out of which only 14.6 million has access to electricity [21]. The northwestern region is the least electrified region in Nigeria. Two states from this region were selected for this study. The selected locations are Kaduna (10.31°N, 7.26°E) and Katsina (12.15°N, 7.30°E). These locations were selected for study because of the high number of rural villages without electricity supply in these states. According to the National Bureau of Statistics household electrification rate survey, the percentage of household without electricity in Kaduna and Katsina States are 46.2% and 68.5%, respectively [21]. Arowolo et al. [22] reported 2998 and 2559 un-electrified clusters for Kaduna and Katsina, respectively. Some of these clusters are presented in **Table 1**. Meteorological data required for this study were not available at the rural villages/clusters scattered across the selected states. Hence, only the data collected at the state capitals by the Nigerian Meteorological Agency were used. The data were collected from the Lagos office of the Nigerian Meteorological Agency. Solar energy potential may vary slightly across different locations in the selected states, but the available global solar radiation at any desired location is adequate for PV application [12]. Ground-measured data can be interpolated between two sites. Nevertheless, the accuracy of interpolation of data decreases with increase in distance between the sites [23].

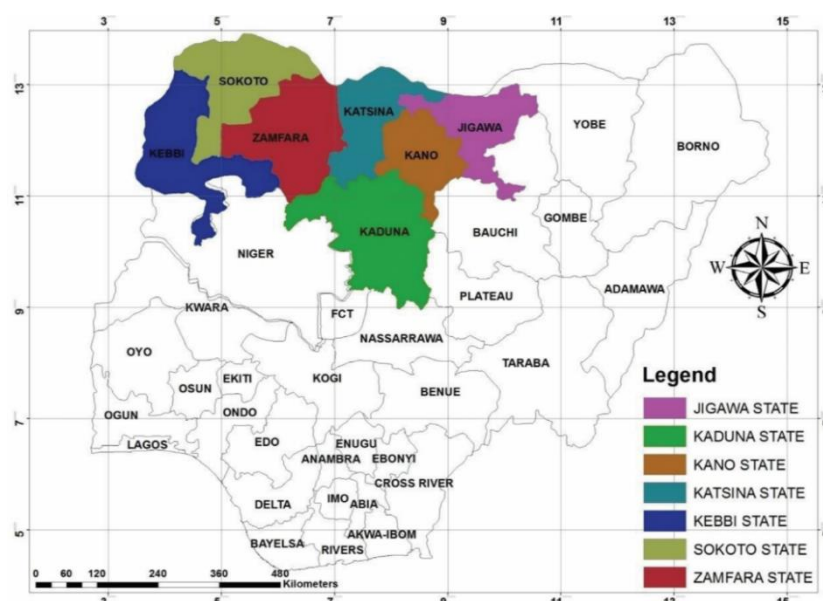


Figure 1 • Map of Nigeria showing the northwestern states.

Table 1 • List of some un-electrified clusters in Kaduna and Katsina states

State	LGA	Ward	Population	Load (MW)
Kaduna	Jema'a	Asso	24,009	8.0
Kaduna	Jaba	Fada	51,119	17.0
Kaduna	Kaura	Kpak	167,398	55.8
Kaduna	Zango-Kataf	Zonkwa	29,730	9.9
Kaduna	Zango-Kataf	Kachia	20,676	6.9
Kaduna	Kuban	Pambegua	31,958	10.7
Katsina	Safana	Rade\B\	53,086	17.7
Katsina	Zango	Rogogo/Cidari	22,099	7.1
Katsina	Daura	Mazoji A	38,975	13.0
Katsina	Mai'Adua	Mai\Adua\B\	29,324	9.8
Katsina	Bakori	Barde/ Kwantakwaran	34,530	11.5
Katsina	Kafur	Kafur	27,300	9.1
Katsina	Dan Musa	Dan Musa A	44,910	15.0
Katsina	Ingawa	Doro	20,409	6.8

Source: Arowolo et al. [22].

2. Methodology

2.1. Simulation of hybrid power systems

In this study, the HOMER software was used for the simulation and analysis of the hybrid off-grid power systems. This software simulates the operation of different power system configurations and ranks them on the basis of their net present cost (NPC; the total installation and operating cost of the system over its lifetime). Three stand-alone power system configurations were examined for each of the two locations in the study area. The choice of the system configurations was based on the available renewable energy resources (solar and wind) in the selected locations. The three stand-alone power system configurations are as follows:

- i wind-PV-battery system
- ii wind-PV-diesel-battery system
- iii diesel system

2.2. Load demand assessment

The HOMER software requires the load profile to be met by the system. Many of the un-electrified clusters in the study locations have less than 100 households. Hence, for the purpose of this study, electrical energy demand for a model community of 100 households was estimated for each of the study locations. This is to ensure uniformity of assessment and to have a basis for comparison between the locations. The daily energy demand (DED) in kilowatt-hour was estimated using the following equation:

$$\text{DED} = \frac{\text{Power rating} \times \text{No. per household} \times \text{Hours use per day} \times \text{No. of household}}{1,000} \quad (1)$$

The electric energy consumption analysis for the model rural community of 100 households is estimated from common appliances used in Nigerian rural households as presented in **Table 2**. The total energy demand of the community was estimated to be 184.1 kWh/day, while the average daily energy demand per home was analysed to be 1.8 kWh.

2.3. System components selection, specification, and cost

The HOMER software also requires the system components (PV panel, wind turbine, generator, battery, and converter) and their respective cost as part of the input parameters for the simulation of the hybrid power system. The specifications of the selected system components, their capital cost, replacement cost, operation and maintenance (O&M) cost, and the expected lifetime are shown in **Table 3**.

HOMER considers all cost in dollars (\$), all calculation involving cost is therefore done with the current exchange rate, and an annual interest rate of 10% is considered. The cost of the solar module, battery, and inverter presented in **Table 2** was collected from a solar products vendor (Inverter Power Option Ltd.) in Minna, Nigeria, while the cost of the wind turbine and generator was retrieved online.

2.4. Wind and solar resources data

The wind and solar resources data presented in **Figure 2** were utilized in the simulation analysis. Clearness index of the selected sites was generated by the HOMER software. The monthly mean wind speed data used were measured at 10 m above the ground level.

Table 2 • Electric consumption analysis for a model rural community of 100 households

Appliance	Power rating (W)	No. per household	Hours of use per day (Hr)	Days of use per week	Daily energy demand (kWh)
TV	90	1	6	7	54.0
Fan	24	3	8	7	57.6
Bulb	18	5	7	7	63.0
Radio	6	1	5	7	3.0
Cell phone	4	3	2	5	2.4
Water pump	400	2/community	2	4	1.6
Clinic equipment	500	Community based	5	5	2.5
Total energy demand of the community (kWh/day)					184.1

Table 3 • System specifications and cost

Component	Model	Size	Capital cost (US\$)	Replacement cost (US\$)	O&M cost (US\$)	Lifetime
PV module	Flame Solar	1 kW	1,050	1,050	0	25 years
Wind turbine	FD4.0-3000	3 kW, 240 V	1,500	1,500	15	15 years
Battery	Luminous	200 Ah, 24 V	355	355	0	5 years
Inverter	Luminous	5 kW	968	968	10	10 years
Generator	HYD 120US	9 kW	3,774	3,774	0.05	20,000 Hours

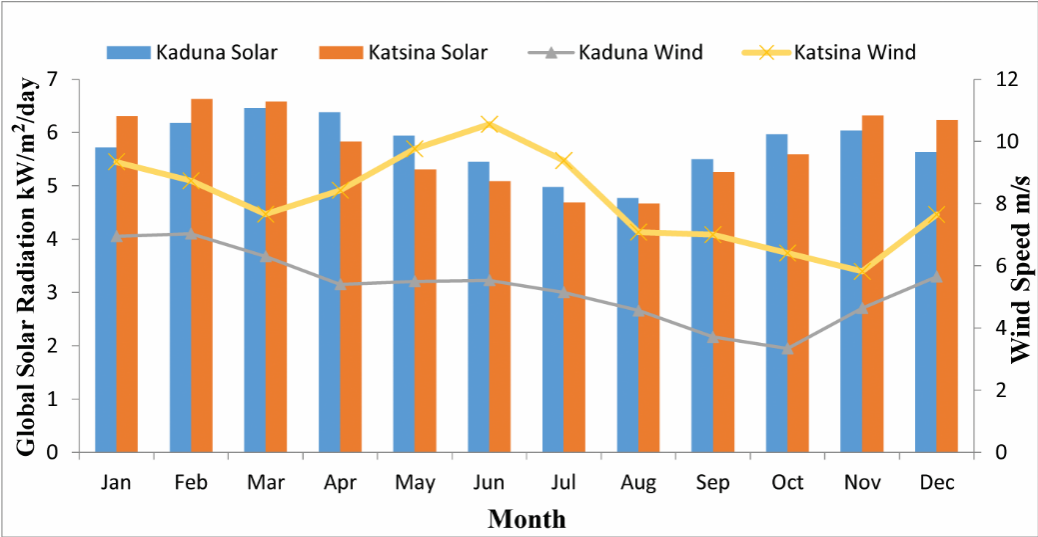


Figure 2 • Monthly mean wind speed and global solar radiation.

3. Results and discussion

3.1. HOMER simulation results

Figure 3 is the schematic diagram of the hybrid system configuration and electric load profiles of the model rural community as displayed in the HOMER software.

A minimum load of 3 kW occurs during the day and a peak load of 13 kW is observed between 20.00 and 22.00 hours (8.0 and

10.0 p.m.) due to the increased lighting load in the homes at this time of the day.

The HOMER simulation and optimization analysis results for Kaduna and Katsina are discussed below. The simulation results showing a list of selected feasible system options and their net present cost (NPC) for the study sites are presented in Tables 4 and 5.



Figure 3 • Hybrid system configuration and electric load profile.

Table 4 • Summary of HOMER optimization results for Kaduna

PV (kW)	No. of wind turbines (3 kW)	Generator (kVA)	Battery (200 Ah)	Converter (kW)	COE (US\$/kWh)	NPC (US\$)	Renewable fraction (%)
15	20	18	25	10	0.203	435,818	55.0
30	–	18	19	10	0.235	503,880	30.2
30	23	27	–	10	0.265	567,889	40.6
30	–	27	–	10	0.297	636,416	16.9
–	–	27	–	–	0.366	785,762	0.0
200	120	–	100	20	0.423	868,999	100

Table 5 • Summary of HOMER optimization results for Katsina

PV (kW)	No. of wind turbines (3 kW)	Generator (kVA)	Battery (200 Ah)	Converter (kW)	COE (US\$/kWh)	NPC (US\$)	Renewable fraction (%)
20	17	9	43	10	0.150	322,091	75.2
15	25	18	–	5	0.158	338,801	66.0
25	45	–	80	15	0.170	349,856	100
30	–	18	19	10	0.235	504,459	30.1
30	–	27	–	10	0.298	638,641	16.6
–	–	27	–	–	0.366	785,762	0

3.2. Kaduna optimization results

Table 4 shows a summary of the optimization results for Kaduna. The optimization results include the different system options generated by HOMER such as PV-wind-diesel-battery, PV-diesel-battery, PV-wind-diesel, PV-diesel, PV-wind-battery, and diesel. The monthly electricity generation by the optimal system (PV-wind (G3)-diesel generator) is shown in **Figure 4**.

In **Table 4**, the least cost system option for the site is the wind-PV-diesel hybrid power system. The optimal hybrid configuration consists of 15 kW PV array, 20 wind turbines of 3 kW each (60 kW), 18 kVA (two 9 kVA) generator, 10 kW inverter, and a

battery bank of twenty-five 200 Ah batteries, with an NPC of US\$435,818. The excess energy generated by the least cost system is accounted by the software to ensure system reliability and can be used to cater for future expansion. A feasible hybrid wind-PV system (i.e., 100 % renewable system) in Kaduna has an NPC of US\$868,999, while a 100% diesel option consisting of three 9 kVA (27 kVA) generators has an NPC of US\$785,762. The high NPC value of the wind-PV hybrid system may be due to the low wind regime in Kaduna, which leads to an increase in the number of the system components as shown in the last row of **Table 4**.

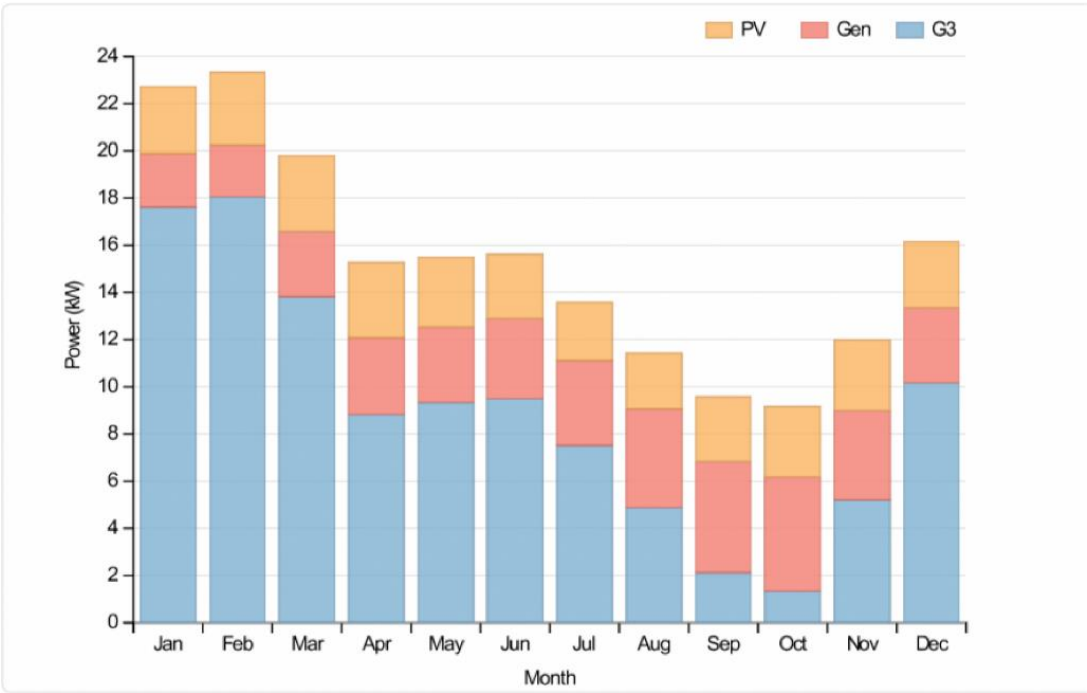


Figure 4 • Monthly share of electricity generation by PV, wind turbine, and generator in Kaduna.

The total electricity generated by the least cost hybrid system is 134,082 kWh/yr, while the AC load being met is 67,018 kWh/yr. The wind turbine produced 78,716 kWh/yr (which is 59% of the total electricity generated), PV array produced 25,185 kWh/yr, and generator produced 30,181 kWh/yr of the total electricity generated. Excess electricity generated by the hybrid system (65,605 kWh/yr) is stored in the battery bank as backup against days with fluctuating solar and wind energy. From **Figure 4**, it can be observed that the wind turbine produced the largest share of electricity throughout the year, except for September and October. This is due to the low wind speed recorded in these months.

3.3. Katsina optimization results

Table 5 shows the summary of optimization results for Katsina. The optimization results include the different system options generated by HOMER such as PV-wind-diesel-battery, PV-diesel-battery, PV-wind-diesel, PV-diesel, PV-wind-battery, and diesel. The monthly electricity generation by the optimal (PV-wind-diesel) system is shown in **Figure 5**.

The optimization results for Katsina presented in **Table 5** reveal that the least cost system option for Katsina is the wind-PV-diesel-battery hybrid option. The optimal hybrid configuration has a 20 kW PV array, 17 wind turbines of 3 kW each (51 kW), a 9 kVA generator, a 10 kW inverter, and a battery bank of forty-three 200 Ah batteries, with an NPC of US\$322,091. A feasible hybrid wind-PV-battery system option (i.e., 100% renewable system) in Katsina has an NPC of US\$349,856, while a 100% diesel option consisting of three 9 kVA (27 kVA) generators has an NPC of US\$785,762.

The total electricity generated by the hybrid system is 221,422 kWh/yr, while the AC load being met is 67,018 kWh/yr. The wind turbine produced 77% (171,485 kWh/yr), the PV array produced 15% (33,336 kWh/yr), and the generator produced 7% (16,600 kWh/yr) of the total electricity generated. Excess electricity generated by the least cost hybrid system (154,404 kWh/yr) is accounted for by the software to ensure system reliability and can be used to cater for future expansion. From **Figure 5**, it can be seen that the diesel generator is only required as backup when the wind turbine/PV cannot meet the energy demand. The electricity produced annually by the wind turbine is far more than the electric load demand. Therefore, only the wind or wind-PV component of the hybrid system can be implemented to meet the community's present and future electrical energy demand. Implementing the wind component of the hybrid system will reduce the NPC of the system from US\$322,091 to US\$176,524, while the wind-PV component has an NPC of US\$197,524.

In summary, the optimal systems for Kaduna and Katsina are hybrid PV-wind-diesel systems with costs of energy (COE) of 0.203 US\$/kWh and 0.150 US\$/kWh, respectively. In both cases, the cost of electricity of the optimal systems are little lower than the cost of electricity from PV and hybrid PV-wind-diesel systems of US\$0.30/kWh and US\$0.22/kWh reported in the World Bank report. However, it is noteworthy to observe that the COEs of the optimal HRRPs for the two locations are in agreement with the values reported in other locations across the world. For instance, findings from the literature on HRRPs indicate a COE of US\$0.11/kWh for a village in the Somali region of Ethiopia [18], US\$0.134/kWh for a village in Malawi [19], US\$0.25/kWh for Fanisau, a remote village of Kano in Nigeria [24], US\$0.373/kWh for a village of Kalakala in the north of Côte d'Ivoire [25], and US\$0.66/kWh for some coastal area in Pekan, Pahang, Malaysia [20].

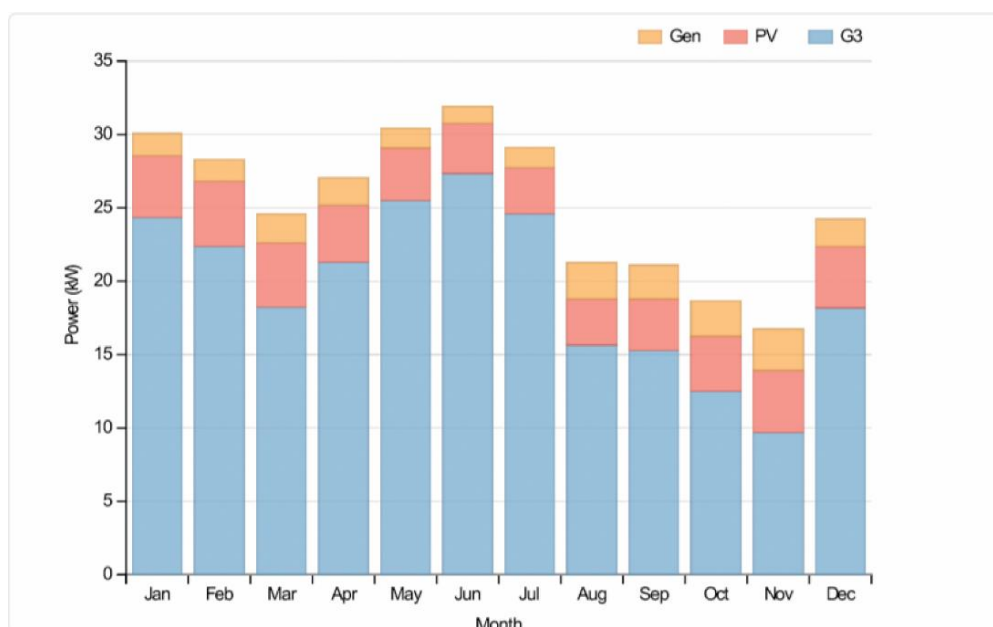


Figure 5 • Monthly share of electricity generation by PV, wind turbine, and generator in Katsina.

The reported cost of energy in this study is higher than the grid tariff of US\$0.09–US\$0.12/kWh in Nigeria. Nevertheless, it is cost effective compared to the cost of grid extension for long distance [5, 6]. Also, the diesel system is not economical compared to the off-grid hybrid PV-wind-diesel systems in this study due to its high maintenance and operating cost. Moreover, the hybrid system options have

great environmental benefits by reducing carbon dioxide emission associated with the diesel system. The adoption of the hybrid system options in the study locations will boost the social, educational, and economic activities in the villages leading to better education, improved healthcare delivery, and enhanced local businesses resulting from reliable power supply.

4. Conclusions

In this study, the electrical load demand of a model community of 100 households with an estimated energy demand of 184.1 kWh/day was simulated using the HOMER software. Two off-grid hybrid power systems and a stand-alone diesel generator are modeled, simulated, and analyzed for Kaduna and Katsina in the northwestern region of Nigeria. The optimization results show that the hybrid wind-PV-diesel power system is sufficient to meet the load demand with the least cost (NPC) in the two locations. The NPC and COE for the optimal system are US\$435,818 and 0.203 US\$/kWh for Kaduna and US\$322,091 and 0.150 US\$/kWh for Katsina, respectively. This indicates that it is more expensive to install a hybrid wind-PV-diesel in Kaduna than Katsina. The reduction in NPC for the optimal system in Katsina is due to the high wind penetration in the hybrid system, as the cost of wind energy per kilowatt is lower than that of PV and diesel generator. In comparison to the grid tariff of US\$0.09–US\$0.12/kWh in Nigeria, the reported cost of energy in this study is slightly higher, but it is cost effective for rural settlements that are far from the existing grid. The adoption of hybrid system options in the study locations will boost the social, educational, and economic activities in the villages. However, the government needs to subsidize the cost of hybrid power system to make investment in such energy system more attractive for citizens, particularly in the rural villages.

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Author contributions

Conceptualization, B.M.O.; methodology, B.M.O.; software, B.M.O.; formal analysis, B.M.O.; writing—original draft preparation, B.M.O.; writing—review and editing, O.D.O.; supervision, O.D.O. Both authors have read and agreed to the published version of the manuscript.

Conflict of interest

The authors declare no conflict of interest.

Data availability statement

Data supporting these findings are available within the article, at <https://doi.org/10.20935/AcadEnergy7345>, or upon request.

Institutional review board statement

Not applicable.

Informed consent statement

Not applicable.

Sample availability

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